

# Project El Tonio

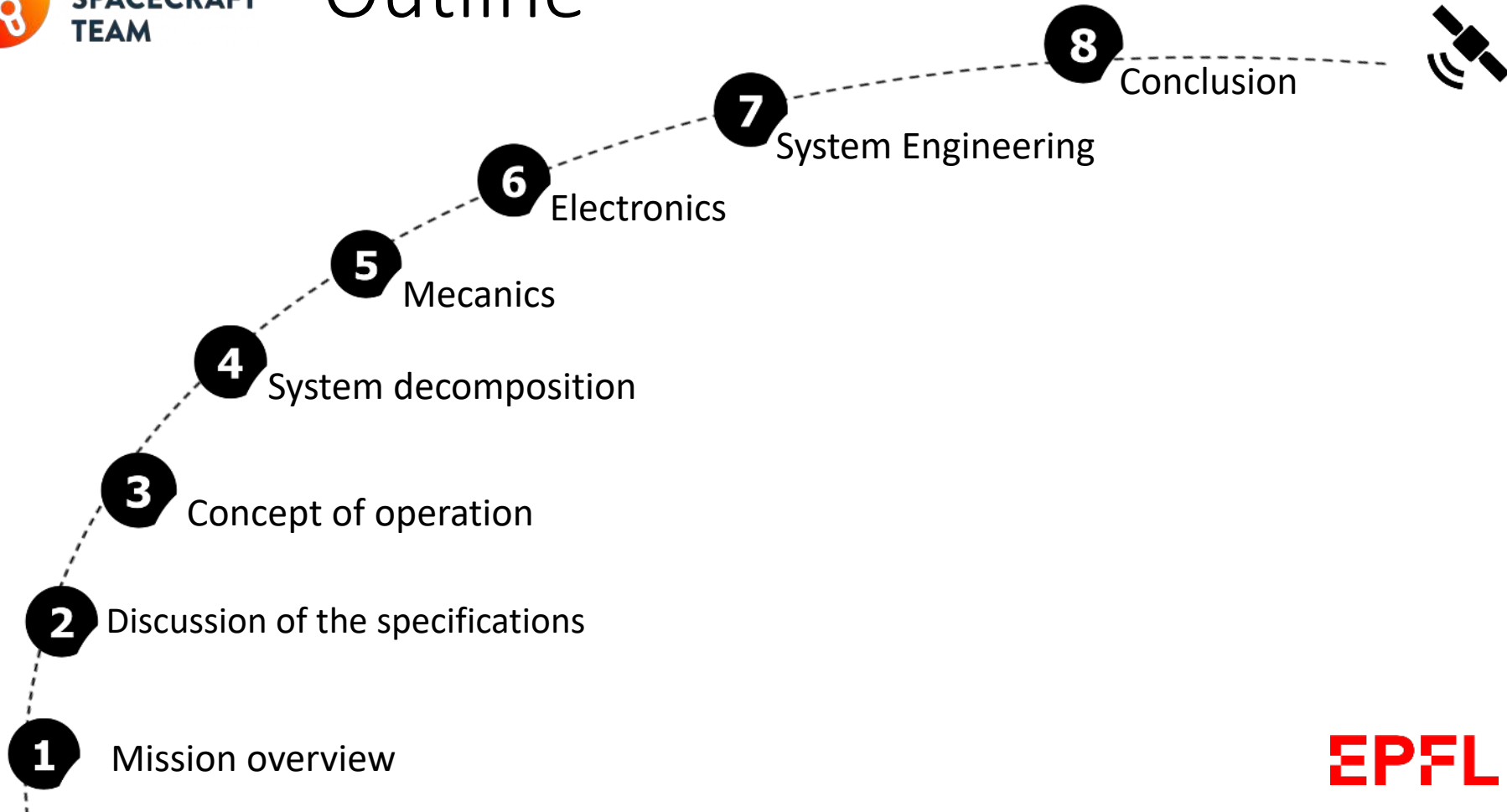
Preliminary Design Review

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# Outline



# Mission overview

The Cansat projet has different objectives:

-  Elaboration of an idea
-  Study of designs and respecting certain features
-  Working as a team
-  Learn to work with a budget
-  Working under time-constraints
-  Practical experience

It is a great opportunity to learn all the way up until the launch !

# Specifications

Interpretation and definition

## General specifications

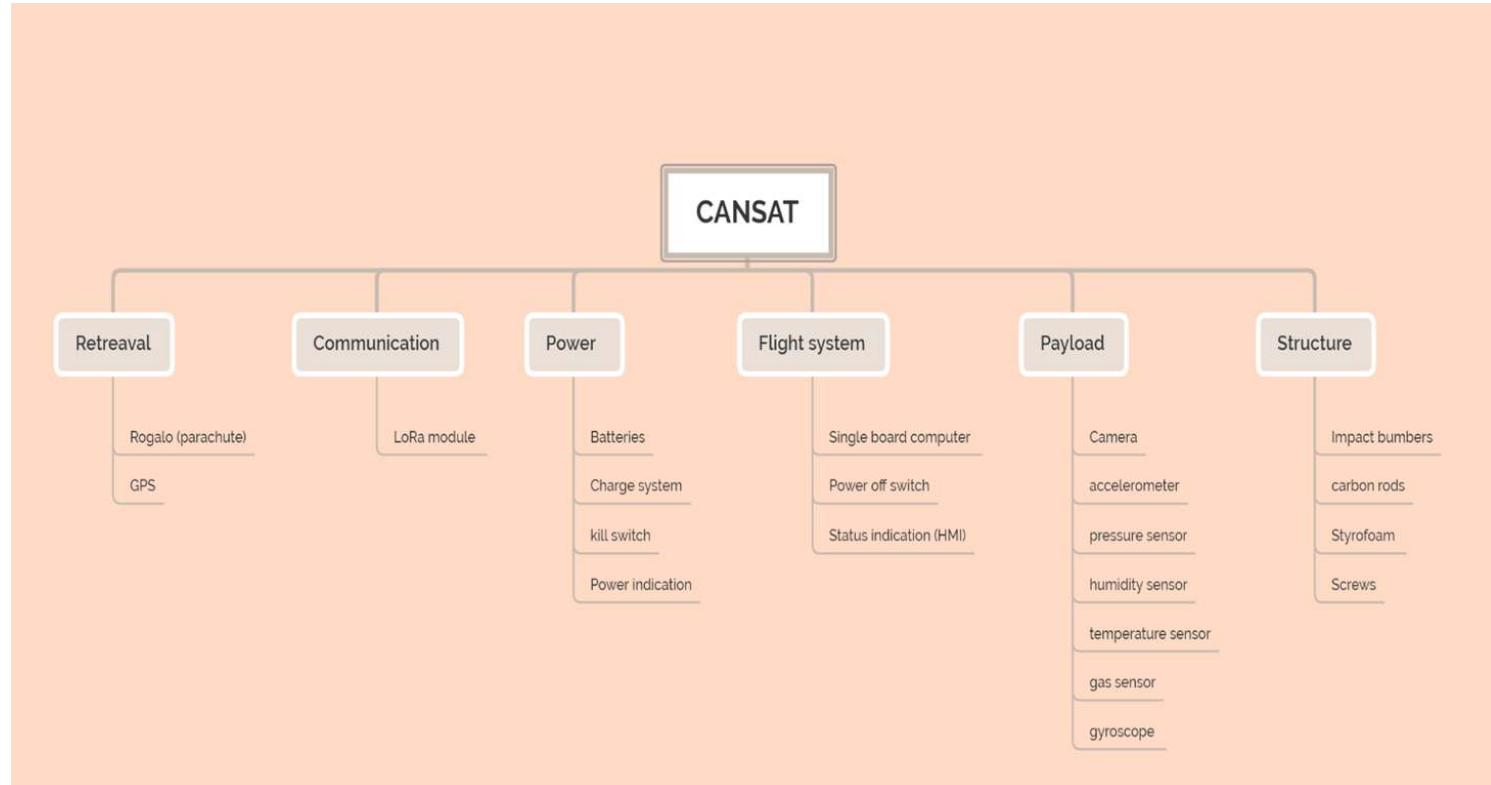
- EPFL spacecraft team's requirements

## Personal specifications

- Implement a paraglider and a guidance system
- We shall film the flight of the Cansat
- Measure of multiple environmental parameters
- Battery monitoring
- Be user friendly HMI
- Easy accessibility
- Professional looking



# Sub-systems decomposition



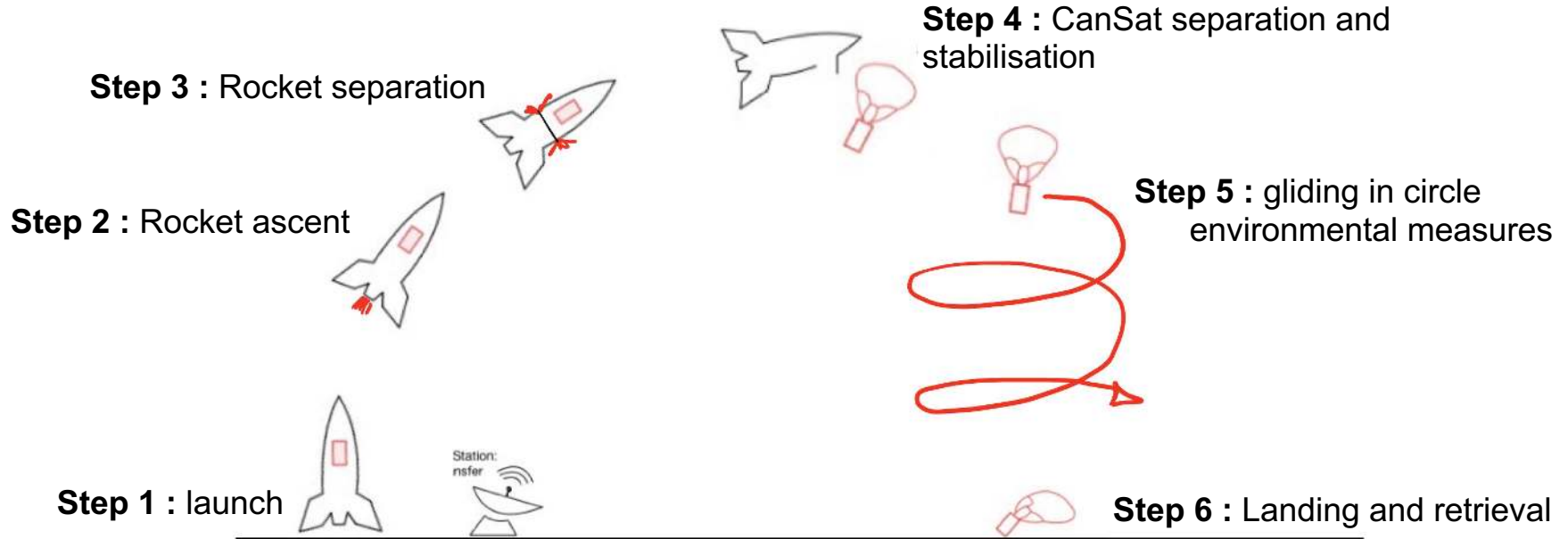


# Sub-systems decomposition



# Concept of operation

## Mission overview



# Mecanics

## Parachute

### First idea :

- Build a paraglider and fly back to launch pad

### Adaptation : Rogallo parachute

- Easy to build
- Horizontal speed component
- Takes less space
- Safe opening





# Mechanics

## Structure

### Idea 1



Batteries and PCB on opposite side

- Easy connection
- Poor stability

PCB and OBC on opposite sides, batteries in the middle

- Better stability during flight

### Idea 2



4 principal rods and 2 horizontal plates



Poor shocks resistance

Lots of screws

# Mechanics

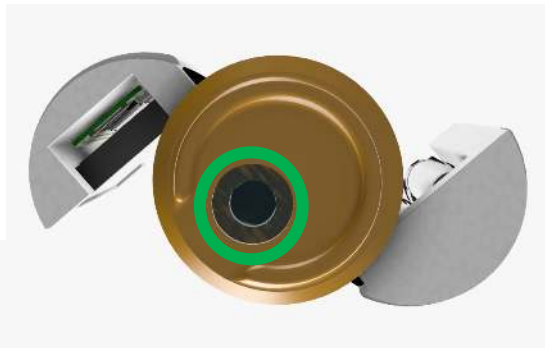
## Structure

*Final idea*

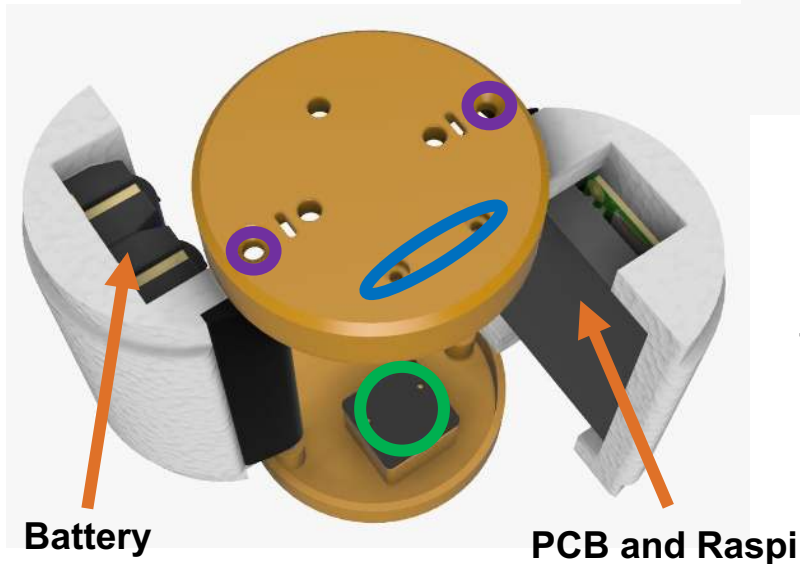


**External bumpers**  
keep the cansat closed  
and act as a skeleton

**Butterfly-like opening**  
Easy and quick access to  
components and batteries



**GPS**, **camera** and  
**parachute** directly  
attached to  
the bumpers



# Mecanics

## Structure

### Weight

| Components            | Weight [g]     |
|-----------------------|----------------|
| Raspberry zero        | 9              |
| PCB                   | 20             |
| GPS                   | 10             |
| Caméra                | 10             |
| Batteries             | 48             |
| Battery holder        | 30             |
| Polystyrene structure | 10             |
| Bumper + rods         | 40             |
| Parachute             | 40             |
| Miscellaneous         | 20             |
| <b>Total</b>          | <b>333 [g]</b> |

### Materials

- Carbon rods  
→ Strong and light
- TPU for the bumpers  
→ Flexible and easy to manuf.
- Polystyrene (exp., extr. or EPP)  
→ Light, cheap, absorbs shocks  
Original

### Construction & manufact.

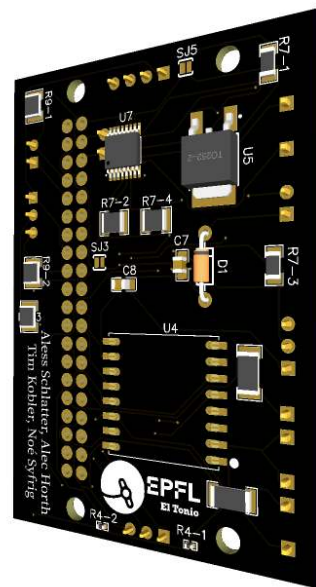
- Machining
- 3D printing  
→ Flex and PLA
- Screws and glue for the rest

# Electronics

## Telemetry

### General criteria of choice

- Temperature range :  $-10^{\circ}\text{C}$  to  $50^{\circ}\text{C}$
- Arduino and python library
- Quality
- Price and availability



**We took our inspiration from Adafruit Sparkfun to choose our sensors and build our own PCB**

A big thanks to Taras Pavil and Michael Linder that have helped us to improve our PCB

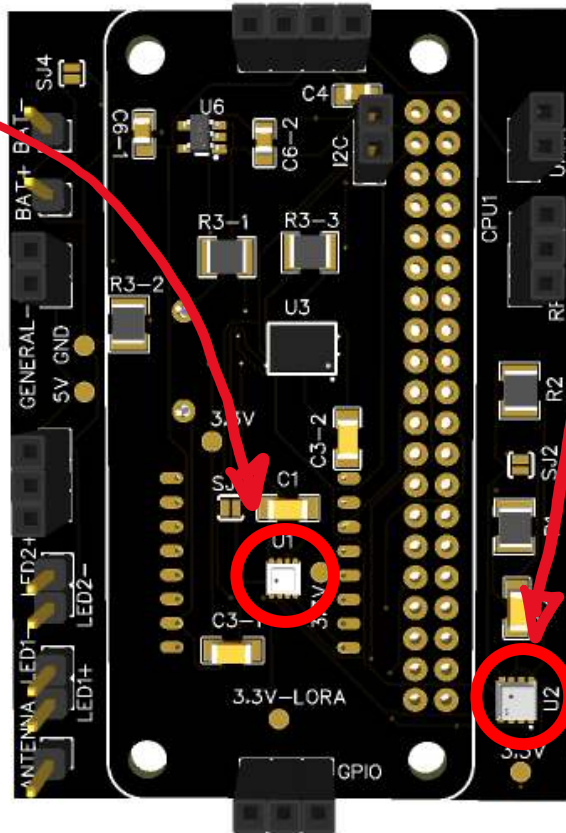
# Electronics

## PCB

### BME280 (temp2)

Humidity (+-3%)  
Pressure (+-1hPa)  
Temperature (+-1°C)

I2C comm.  
3,3V alim.



### BME688 (temp1)

Humidity (+-3%)  
Pressure (+-1hPa)  
Temperature (+-1°C)  
VOC gaz

I2C comm.  
3,3V alim.

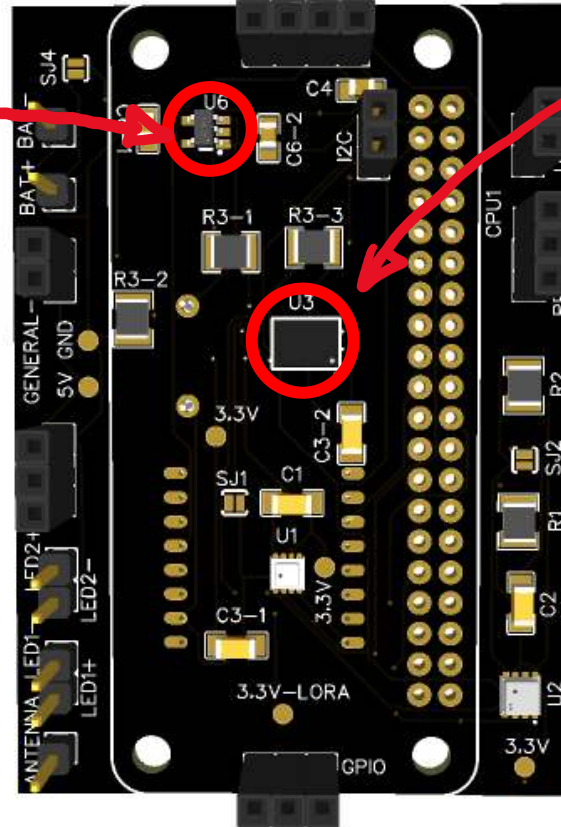
# Electronics

## PCB

**3,3V high Amper**

Alows to **properly power** the **communication module** and the **GPS**.

Convert 5V to 3,3V with a higher amper capacity



**BNO055 (accelo)**

Accelerometer  
Gyroscope  
Magnetometer

I2C comm.  
3,3V alim.

# Electronics

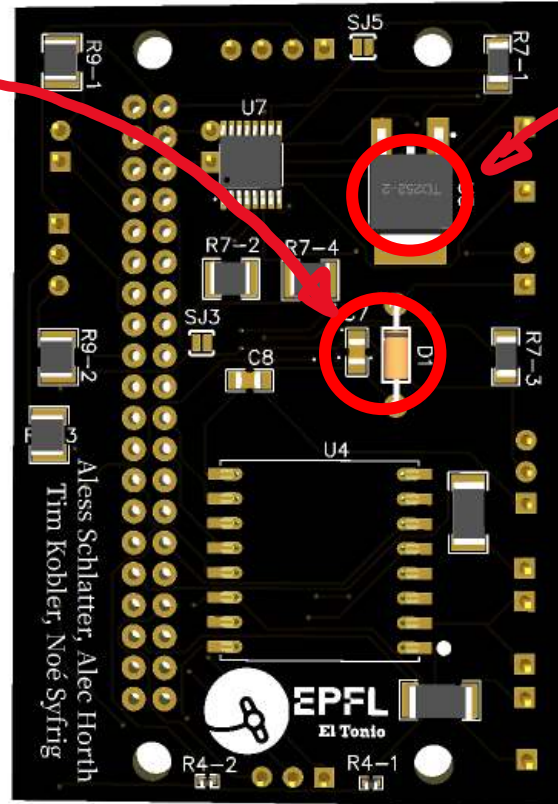
## PCB

Raspi protection

Zener Diode  
1u capacitor

6,6 V to 5V converter

Allows to take take  
down the 6,6V battery  
to stable 5V to power  
the Raspi trough the  
GPIOs

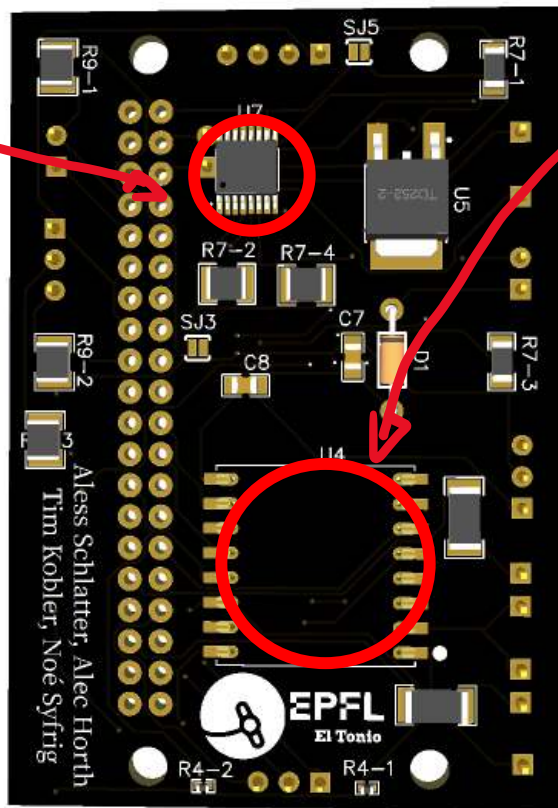


# Electronics

## PCB

### INA260

Voltage measure and  
current measure of the  
battery



### LoRa comm. module

868 MHz

Data rate : 300kbps

Power output : 20dBm

Sensitivity : -148dBm

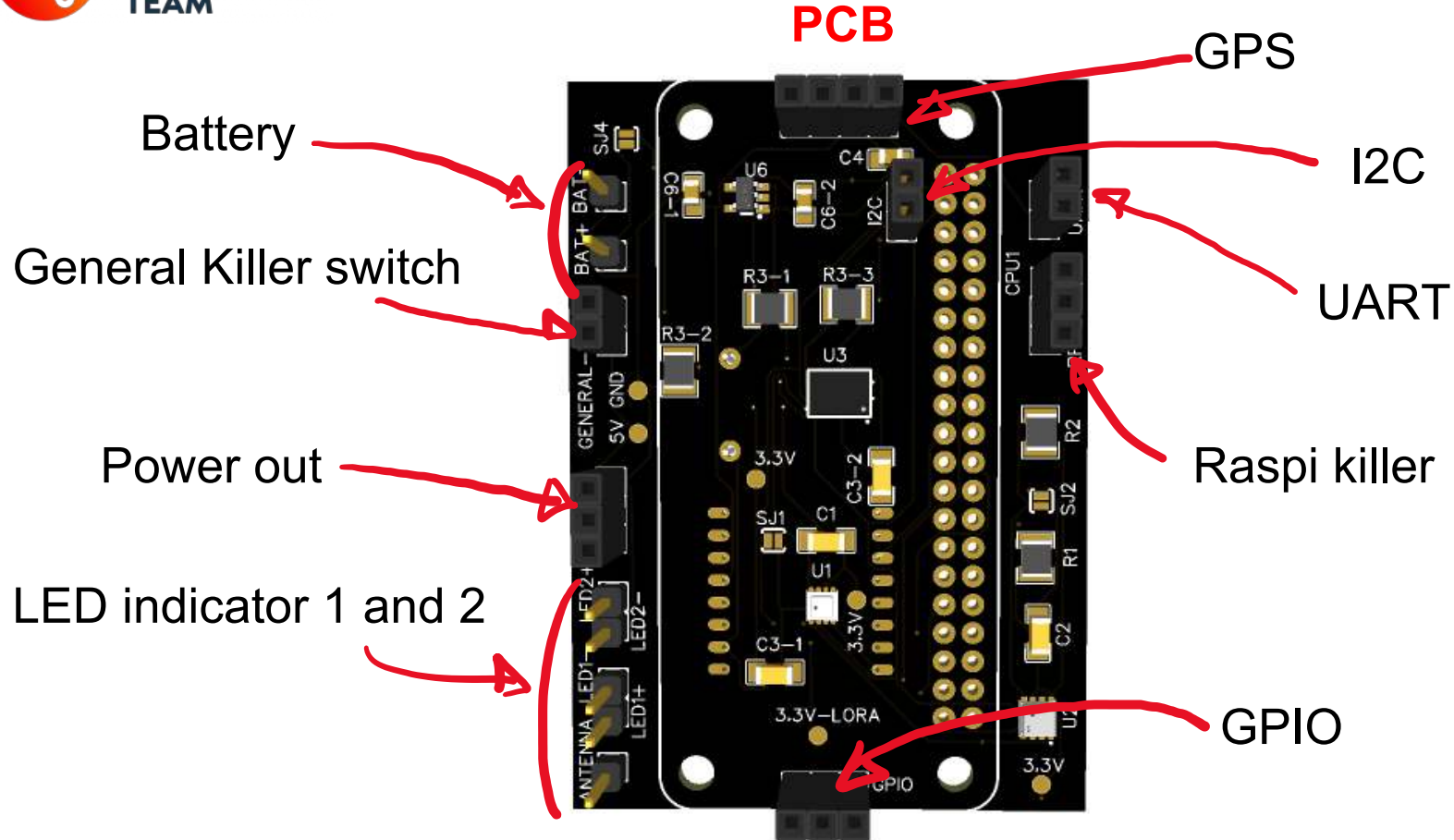
SPI comm.

3,3V





# Electronics



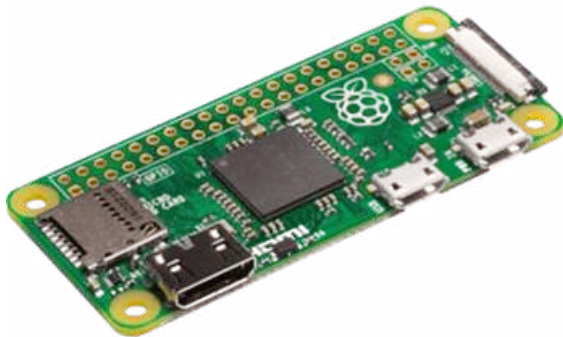
# Electronics

## Microcontroller

### Raspberry pi zero

Main reasons for our choice:

- Supports a relatively good quality camera
- Integrated Sd-Card



#### Short description

- |                                                                                                                                                                            |                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• 1GHz single-core CPU</li><li>• 512MB RAM</li><li>• Mini HDMI port</li><li>• Micro USB OTG port</li><li>• Micro USB power</li></ul> | <ul style="list-style-type: none"><li>• HAT-compatible 40-pin header</li><li>• Composite video and reset headers</li><li>• CSI camera connector (v1.3 only)</li></ul> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# Electronics

## Power supply

| Rechargeable Battery type                    | Lead      | Alkaline    | Nickel    |           |           | Lithium     |             |             |             |
|----------------------------------------------|-----------|-------------|-----------|-----------|-----------|-------------|-------------|-------------|-------------|
| Criteria                                     | Lead-acid | RAM         | Ni-Fe     | Ni-Cd     | Ni-Mh     | Li-Ti       | Li-Fe       | Li-Co       | Li-Mn       |
| Masse energy density [Wh/kg]                 | 35,00     | 80,00       | 21,00     | 30,00     | 100,00    | 110,00      | 160,00      | 195,00      | 150,00      |
| Volumic energy density [Wh/dm <sup>3</sup> ] | 67,50     | 160,00      | 125,00    | 100,00    | 400,00    | 177,00      | 333,00      | 560,00      | 420,00      |
| Voltage per cell [V]                         | 2,10      | 1,57        | 1,20      | 1,20      | 1,20      | 2,30        | 3,20        | 3,70        | 3,90        |
| Cutoff voltage [V]                           | 1,75      | 0,90        | 0,75      | 0,90      | 0,90      | 1,80        | 2,00        | 2,50        | 2,50        |
| Market disponibility                         | ---       | +++         | --        | +++       | +         | -           | ++          | +++         | +           |
| Heat résistance [°C]                         | -40 to 49 | -18 to 55   | -40 to 40 | -60 to 65 | -30 to 75 | -20 to 60   | -20 to 60   | -20 to 60   | -20 to 60   |
| Vibration & impact résistance                | No        | Yes         | Yes       | Yes       | Yes       | Yes         | Yes         | Yes         | Yes         |
| Maintenance                                  | +++       | --          | -         | -         | -         | +           | +           | +           | +           |
| Safety                                       | --        | +++         | ++        | +         | +++       | -           | ++          | --          | -           |
| Price/cell (approximation)                   | ++        | +++         | -         | ++        | +++       | ---         | +           | +           | +           |
| Durability [life cycle]                      | 1500      | 2000 - 3000 | 700-1000  | 700-1000  | 700-1000  | 3000 - 5000 | 3000 - 5000 | 3000 - 5000 | 3000 - 5000 |

Electronics

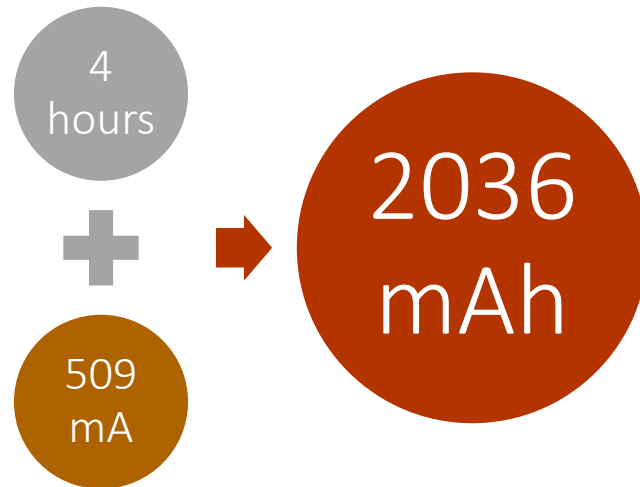
Power supply

| Lead      | Alkaline | Nickel |       |       | Lithium |       |       | Weight coefficient [%] |       |
|-----------|----------|--------|-------|-------|---------|-------|-------|------------------------|-------|
| Lead-acid | RAM      | Ni-Fe  | Ni-Cd | Ni-Mh | Li-Ti   | Li-Fe | Li-Co | Li-Mn                  |       |
| 0,03      | 0,09     | 0,03   | 0,06  | 0,09  | 0,09    | 0,12  | 0,15  | 0,12                   | 15,00 |
| 0,01      | 0,02     | 0,02   | 0,02  | 0,08  | 0,02    | 0,06  | 0,10  | 0,08                   | 10,00 |
| 0,02      | 0,01     | 0,01   | 0,01  | 0,01  | 0,03    | 0,04  | 0,04  | 0,04                   | 5,00  |
| 0,03      | 0,04     | 0,04   | 0,04  | 0,04  | 0,04    | 0,03  | 0,03  | 0,03                   | 5,00  |
| 0,00      | 0,20     | 0,04   | 0,20  | 0,12  | 0,08    | 0,16  | 0,16  | 0,12                   | 20,00 |
| 0,02      | 0,02     | 0,03   | 0,05  | 0,05  | 0,04    | 0,04  | 0,03  | 0,04                   | 5,00  |
| 0,03      | 0,05     | 0,05   | 0,05  | 0,05  | 0,05    | 0,05  | 0,03  | 0,05                   | 5,00  |
| 0,10      | 0,02     | 0,04   | 0,04  | 0,04  | 0,06    | 0,06  | 0,06  | 0,06                   | 10,00 |
| 0,03      | 0,15     | 0,12   | 0,09  | 0,15  | 0,06    | 0,12  | 0,08  | 0,06                   | 15,00 |
| 0,04      | 0,05     | 0,02   | 0,04  | 0,05  | 0,00    | 0,03  | 0,03  | 0,03                   | 5,00  |
| 0,01      | 0,02     | 0,04   | 0,04  | 0,04  | 0,05    | 0,05  | 0,03  | 0,05                   | 5     |
| 0,31      | 0,67     | 0,44   | 0,64  | 0,72  | 0,52    | 0,76  | 0,72  | 0,68                   |       |

# Electronics

## Power supply - Consumption

| Component         | Power consumption |
|-------------------|-------------------|
| Raspberry pi zero | 100 mA à 370 mA   |
| RFM96W (LoRa)     | 30 à 100 mA       |
| BME688 (Pression) | 1 à 4 mA          |
| BME280 (Pression) | 0.4 mA            |
| GPS               | 29 à 34 mA        |
| BNO055            | 80 $\mu$ A        |
| LIS331            | 370 $\mu$ A       |
| INA260            | 310 $\mu$ A       |
| Maximum total     | 509 mA            |





# Electronics

## Power supply batteries Ansmann

2 x 3,6 V Output

3600 mAh Capacity

IEC 18650 – Aluminium casing

Rechargeable by micro-USB 5V

Short-circuit protection



# Electronics

## Power supply features

### Voltage regulation

- L7805
- 0..35V -->5V
- 0.5A

### Power off

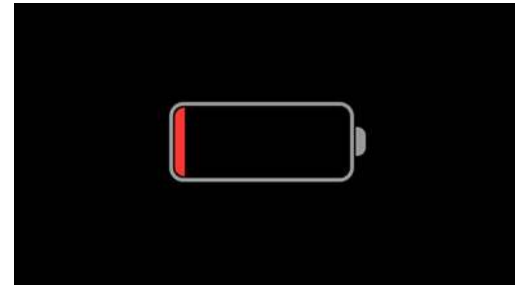
- Kill off switch
- Raspi shutdown switch

### Power monitoring

- Tension and current monitoring
- Coulomb counter

### Batteries housing

- IEC 18650
- Easily replaceable



# Electronics

## Ground station

### General Concept

- Lora module identical to the one on the CanSat
- Controlled by an Arduino
- Communication with the computer through an USB cable and to the serial monitor.
- Non-directional antenna for the reception

Note: We won't create a PCB for this, a simple "Perfboard" should suffice

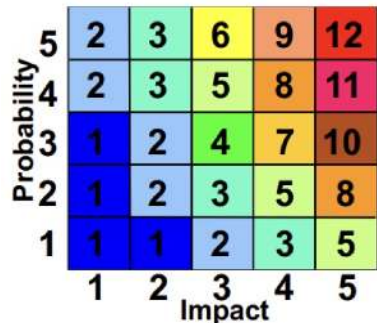


If we have the opportunity to find an "all in one" usb-dongle, we will preferably use this



# System engineering

## Risk analysis



| Level | Technical Criteria                                  | Eventual Solution                                                                                      |
|-------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| 9     | Parachute deployment                                | Rogallo parachute                                                                                      |
| 6     | Retrieval of the CanSat                             | GPS, Bright colours for the CanSat                                                                     |
| 4     | Environmental sensors damaged during launch         | BME redundancy (two different types)                                                                   |
| 5     | Structure integrity                                 | Flexible and light material                                                                            |
| 7     | Battery catching fire                               | Battery monitoring, Li-ion (aluminium casing)                                                          |
| 6     | Telecom range too short                             | High range LoRa comm. Module, and Sd-card for later recovery of the data and post-processing           |
| 3     | Ground receiver application crashes                 | Backup text file with the received data, quick restart functionality                                   |
| 5     | on board storage gets corrupted/ damaged on landing | telecom monitoring and storage on computer                                                             |
| 7     | battery does not last 4h                            | Enable low power modes (GPS, ...), transmit only essential data                                        |
| 2     | Sensor disconnects                                  | ensure connections are robust, ensure software can recover from unresponsive io, redundancy of sensors |

# System engineering

## Budget estimation

| Category          | Estimated price |
|-------------------|-----------------|
| Structure         | 30 CHF          |
| Parachute         | 10 CHF          |
| PCB manufacturing | 20 CHF          |
| PCB components    | 80 CHF          |
| Other electronics | 40 CHF          |
| Testing           | 20 CHF          |
| Shipping          | 50 CHF          |
| <b>TOTAL</b>      | <b>250 CHF</b>  |

# System engineering

## Requirement compliance

General

- **Structure requirements:**
  - 75x110 mm so it leaves 40 mm for the parachute
  - Main structure in polystyrene so it's pretty light (~330g)
  - Flexible bumpers and structure to protect from shocks
- **Feature requirements:**
  - Batteries and electronics easily accessible thanks to butterfly opening
  - 2 switches to control power supply
  - A customized PCB with multiple environmental sensors
  - A LoRa module and a GPS allows recovery and communication
  - Primary mission: Camera & PCB design
  - Secondary missions: pressure, temperature, telecommunication, position in space
  - Rogallo parachute for recovery (4m/s fall rate)
- **Logistic requirements :**
  - Budget: 250.-
  - No operation is irreversible so multiple launches are possible

# System engineering

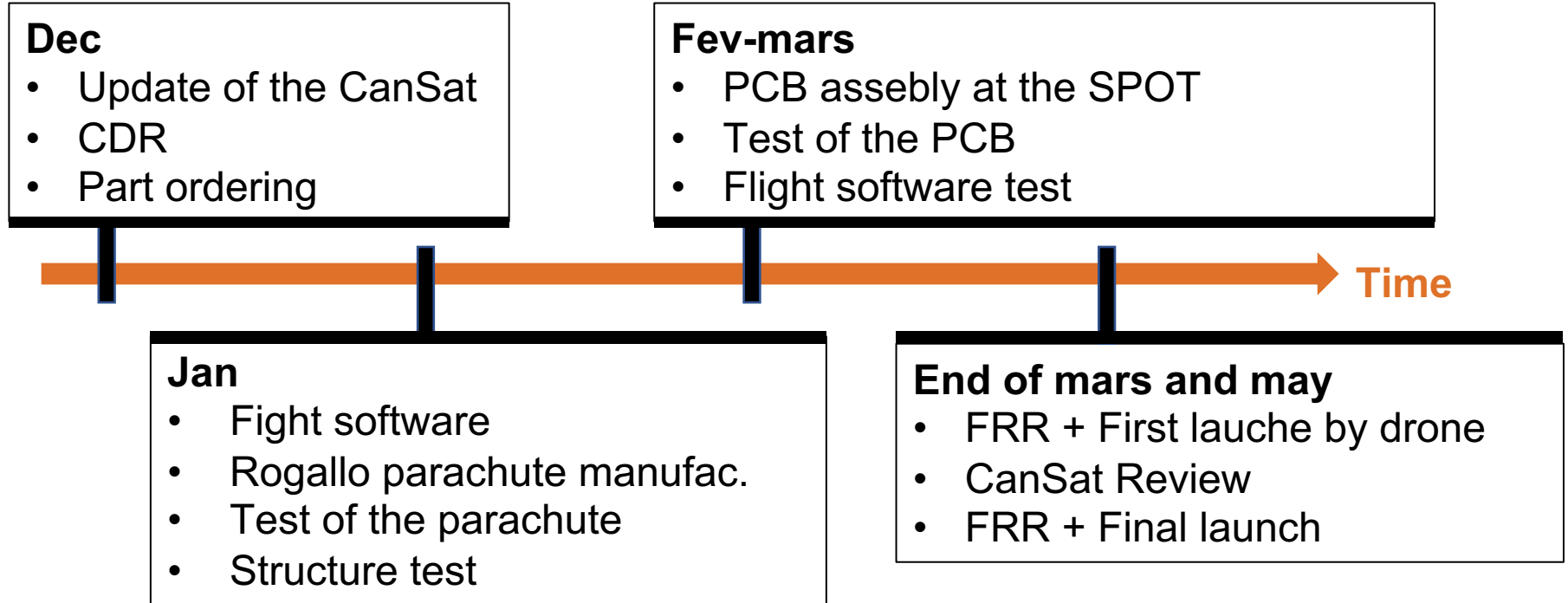
## Requirement compliance

*Our requirements*

- We didn't succeed to implement a paraglider
- We implemented a camera
- Our customized PCB has multiple sensors for environmental parameters and power regulation
- We implemented switches and LEDs to have a user friendly HMI
- Our structure with butterfly opening and polystyrene casing allow an easy accessibility

# System engineering

## Schedule



# System engineering

## Testing plan

GPS test (accuracy)

Communication test (distance and orientation, data rate)

Sensors test (temperature, pression, acceleration, giro)

Longevity test of the power system

Parachute test and steering calibration

Material and structure test (rupture, solidity, etc...)

Camera and OBC test

# Conclusion

- We succeeded to respect most of our requirements (Only the paraglider wasn't implemented)
- On our PCB we have extra pins for future extra sensors and unexpected changes
- We learned how to combine EPFL related work and the organisation of our spare time as a group
- Looking forward for the next step!



If you have any questions, we are happy to answer them!